

HIBO EnergyPlus Building Model

Model Inputs and Simulation Results

Juhyun Bak | University of Nebraska–Lincoln
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1. Model Overview

The Human-Centered Integrated Building Operation (HIBO) laboratory in Omaha, Nebraska, is modeled with three directly served thermal zones, a separate MEP zone, and a common return plenum. Lab A and Lab B are separate zones; the Lobby, Restroom, and Control rooms form one combined zone. One common air loop provides two-speed direct-expansion (DX) cooling and central electric heat, with a variable-air-volume (VAV) terminal and electric reheater for each served zone. [1,3]

Table 1-1. HIBO model summary

Characteristic	Value / representation
Modeled total floor area	67.782 m ² , including MEP; derived from floor polygons
Directly served floor area	61.325 m ² across three main zones
Stories / thermal zones	One occupied level plus plenum / five thermal zones
Typical occupied ceiling height	3.048 m; restroom portion 2.7432 m
Main / rear roof elevations	4.7244–4.5212 m / 4.0640–3.9624 m
Exterior walls	Metal-panel and CMU-veneer assemblies
Roof / floor	EPDM with insulation above metal deck / ground-contact concrete slab
Glazing	SN54 simple-glazing reference externally; 6.35 mm clear glass internally
Air system	One common VAV air loop with central DX cooling and electric heat
Terminal systems	Three VAV terminals, each with electric reheat
Shading / internal mass	Five parapet shading polygons / four central-zone mass objects

Airflows and most coil capacities remain Autosize; the central electric heater is specified at 15 kW. The model combines drawing-based geometry, reference material properties, and operating assumptions; it has not been field-calibrated. [1,3,4]

2. Simulation Settings and Weather

2.1 Simulation Settings

The input specifies EnergyPlus version 22.2 and the supplied output identifies EnergyPlus 22.2.0-c249759bad. Zone and system sizing are enabled, plant sizing is disabled, and the annual weather-file run is enabled. Although separate design-day reporting is disabled, the seven SizingPeriod:DesignDay objects remain sizing inputs. [1,5]

Table 2-1. Simulation settings

Setting	Input	IDF object
EnergyPlus version	22.2.0 (IDF version 22.2)	Version
Run period	January 1–December 31; year fields blank	RunPeriod
Start day / calendar	January 1 = Monday; weather-file holidays and daylight saving disabled	RunPeriod
Weather-file rain / snow indicators	Yes / Yes	RunPeriod
Zone / system / plant sizing	Yes / Yes / No	SimulationControl
Separate sizing-period simulation	No	SimulationControl
Weather-file annual simulation	Yes	SimulationControl
HVAC sizing simulation	No; maximum passes field = 3	SimulationControl
Zone timestep	6 per hour (10 minutes)	Timestep
Minimum HVAC timestep	1 minute	ConvergenceLimits
Maximum HVAC iterations	100	ConvergenceLimits
Plant iteration limits	Minimum 2; maximum 8	ConvergenceLimits
Warmup days	Minimum 6; maximum 25	Building
Loads / temperature convergence tolerance	0.04 / 0.4°C	Building
Envelope heat balance	ConductionTransferFunction	HeatBalanceAlgorithm
Surface-temperature upper limit	200°C	HeatBalanceAlgorithm
Inside / outside convection	TARP / TARP	SurfaceConvectionAlgorithm:*
Solar distribution	FullExterior	Building
Shadow calculation	PolygonClipping; SutherlandHodgman	ShadowCalculation
Shadow update	Periodic; every 20 days	ShadowCalculation
Shadow overlap maximum	15,000 figures	ShadowCalculation
Diffuse-sky treatment	SimpleSkyDiffuseModeling	ShadowCalculation

Source: [1, L20–L95, L323–L336]. The shadow-calculation Pixel Counting Resolution is 512 although the selected method is PolygonClipping; all three external-shadow/self-shading disable flags are No.

2.2 Weather File and Site

Table 2-2. Weather and site inputs

Input	Value	Source
EPW	USA_NE_Omaha-Eppley.Airfield.725500_TMY3.epw	Supplied file [2]
Station / data set	Omaha Eppley Airfield, NE, USA / TMY3	EPW LOCATION header
WMO / latitude / longitude	725500 / 41.32° / -95.90°	EPW LOCATION; Site:Location
Time zone / elevation	UTC-6 / 299 m	EPW LOCATION; Site:Location
Weather records	8,760 hourly records	Counted in supplied EPW
Header data period	January 1–December 31; Sunday	EPW DATA PERIODS
Simulation calendar	January 1 = Monday; year fields blank	RunPeriod in IDF
Holidays / daylight saving	Disabled for the annual run	RunPeriod in IDF
Data provenance	NREL TMY Data Set (2008); generally 1973–2005 period of record	EPW COMMENTS 1

The RunPeriod calendar is the model input governing day-type schedules; it is not replaced here by the EPW header's Sunday start-day label. TMY monthly records retain their source-year values. The filename date identifies the model snapshot, not the meteorological year. The weather file is selected directly in EP-Launch. [1,2]

2.3 Sizing Design Days

Table 2-3. Sizing design-day inputs

Condition	Date	Max dry bulb [°C]	Daily range [°C]	Humidity input	Wind [m/s / °]	Source
Ann Clg .4% Condns DB=>MWB	7/21	35	11	Wetbulb 24.5°C	6.1 / 180	[1, L110–L138]
Ann Clg .4% Condns DP=>MDB	7/21	29.7	11	Dewpoint 24.5°C	6.1 / 180	[1, L140–L168]
Ann Clg .4% Condns Enth=>MDB	7/21	32	11	Enthalpy 82600 J/kg	6.1 / 180	[1, L170–L198]
Ann Clg .4% Condns WB=>MDB	7/21	32	11	Wetbulb 26.1°C	6.1 / 180	[1, L200–L228]
Ann Htg 99.6% Condns DB	1/21	-21.3	0	Wetbulb -21.3°C	4.5 / 340	[1, L230–L258]
Ann Htg Wind 99.6% Condns WS=>MCDB	1/21	-8.6	0	Wetbulb -8.6°C	13.8 / 340	[1, L260–L288]
Ann Hum_n 99.6% Condns DP=>MCDB	1/21	-20.6	0	Dewpoint -25.4°C	4.5 / 340	[1, L290–L318]

All seven design days use 97,784 Pa and DefaultMultipliers; rain, snow, and daylight saving are No. The four summer records use ASHRAETau with beam/diffuse optical depths 0.421 / 2.196. The three winter records use ASHRAEClearSky and sky clearness 0. The remaining blank solar and humidity schedules remain empty. [1, L110–L318]

3. Geometry and Zoning

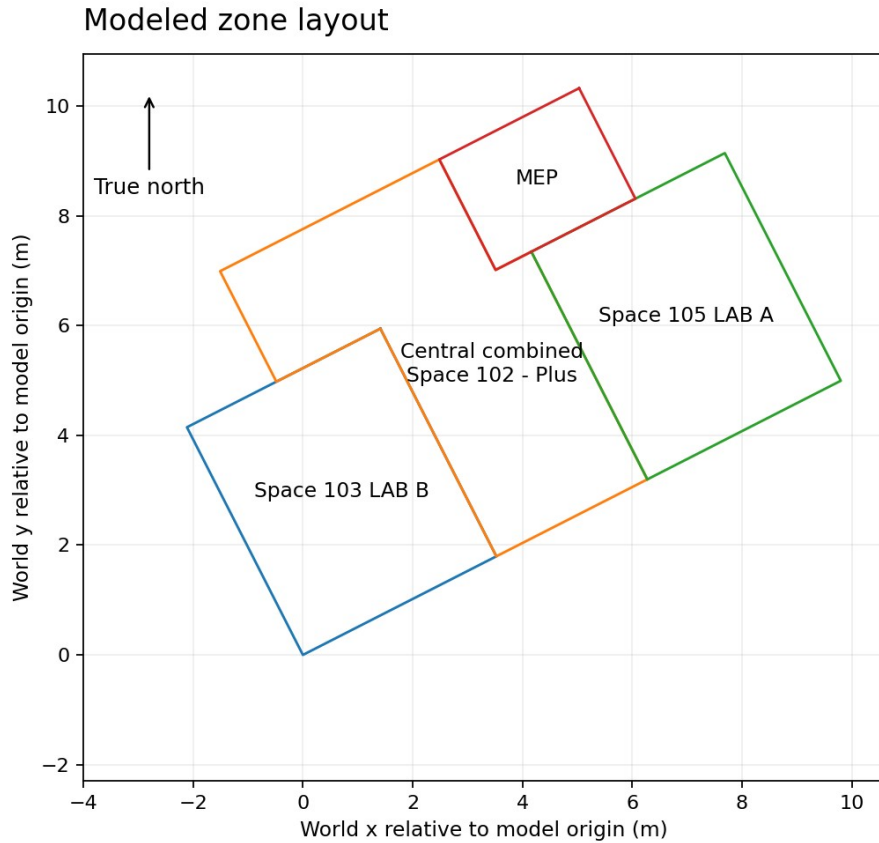


Figure 3-1. Modeled zone layout derived from the current IDF floor vertices. The common plenum is above the occupied plan. [1]

Table 3-1. Zone and architectural-room correspondence

IDF zone	Drawing room(s)	Floor/footprint area [m ²]	Volume [m ³]	Representation
Thermal Zone: Space 103 LAB B	Lab B / room 103	18.414	56.13	Dedicated VAV + reheat
Thermal Zone: Space 102 - Plus	Lobby 101 + Restroom 102 + Control 104	24.497	72.81	One combined zone; VAV + reheat
Thermal Zone: Space 105 LAB A	Lab A / room 105	18.414	56.13	Dedicated VAV + reheat
MEP	MEP / room 106	6.457	19.68	No dedicated terminal
Thermal Zone: Plenum	Common return plenum	67.782	98.50	Above the rooms; not counted in floor-area total

Lab B is modeled as Thermal Zone: Space 103 LAB B, and Lab A as Thermal Zone: Space 105 LAB A. Lobby 101, Restroom 102, and Control 104 share the central zone's simulated air temperature. The plenum footprint is not added again to building floor area. The MEP room has no dedicated air terminal. Model polygon area is not the same quantity as the architectural cover sheet's gross building area. [1,3]

Table 3-2. Zone extents and enclosed volumes calculated from surface vertices

Zone	Model-coordinate extent x × y [m]	Vertical extent [m]	Modeled volume [m ³]
MEP	6.312–9.171 × 4.654–6.913	0.000–3.048	19.682
Thermal Zone: Plenum	0.000–11.000 × 0.000–6.913	2.743–4.724	98.495
Thermal Zone: Space 103 LAB B	0.000–3.956 × 0.000–4.654	0.000–3.048	56.125
Thermal Zone: Space 102 - Plus	1.829–7.044 × 0.000–6.913	0.000–3.048	72.806
Thermal Zone: Space 105 LAB A	7.044–11.000 × 0.000–4.654	0.000–3.048	56.125

The central-zone extent is a bounding rectangle, not a statement that the zone itself is rectangular. Exact vertices, surface assignments, and reciprocal adjacent-surface names are supplied in the IDF and the complete input register. [1]

Table 3-3. Geometry conventions and boundaries

Geometry input	Value / treatment
Coordinate system	Relative; counterclockwise vertices; UpperLeftCorner
Zone origins / zone rotations	All zero
Building North Axis	333°
Main occupied ceiling	3.048 m; restroom portion 2.7432 m
Main roof elevations	4.7244–4.5212 m
Rear roof elevations	4.0640–3.9624 m
Building surfaces	67 BuildingSurface:Detailed objects
Openings and opaque opening pieces	108 FenestrationSurface:Detailed objects; includes doors, frames, mullions and spandrels
Shading	5 Shading:Building:Detailed parapet objects
Internal mass	4 InternalMass objects in the combined central zone
Outside surfaces	Outdoors boundary
Ground-contact slabs	Ground boundary; Section 4.3 temperatures
Interior partitions / ceilings	Paired Surface boundaries to adjacent zone or plenum

The listed zone origins and zone rotations are zero. Surface and opening names are retained from the IDF. Outdoors, Ground, and paired Surface boundaries are kept distinct. Ground-contact boundary temperatures are given in Section 4.3; construction assignments are given in Section 4. [1]

3.1 Openings, Frames, and Shading

The 108 FenestrationSurface:Detailed objects include transparent openings and explicit opaque opening pieces. They should not be counted as 108 transparent windows. Some exterior and interior frame strips, door pieces, and spandrels are represented as opaque Door-type subsurfaces. The two frame-and-divider objects are used for the principal south vision windows. [1]

Table 3-4. Principal south vision window polygons

Window object	Vertex polygon width × height [m]	Frame/divider object	Source
HIBO CWC LabB Vision	3.5560 × 2.5400	HIBO CWC Vision Frame	[1, L3194–L3215]
HIBO CWB Control Vision	1.1176 × 2.5400	HIBO CWB Vision Frame	[1, L3217–L3238]
HIBO CWC LabA Vision	3.5560 × 2.5400	HIBO CWC Vision Frame	[1, L3240–L3261]

The polygon dimensions are not interchangeable with clear-glass area or frame-inclusive opening area. The glazing, explicit frame-strip constructions, and FrameAndDivider inputs are documented separately in Section 4.5. The five Shading:Building:Detailed objects represent the South, West, East, Northwest Shoulder, and Northeast Shoulder upper parapets. Their transmittance-schedule fields are blank; the exact vertices are retained in the complete register. [1]

3.2 Internal Partitions and Same-Zone Mass

Interzone partitions and ceilings use paired Surface boundaries. Partitions and doors separating rooms that have been combined into the central thermal zone are instead represented by InternalMass objects with half-layer constructions and the entered exposed areas. These mass areas are not additional floor area. No explicit door-opening airflow is assigned. [1]

Table 3-5. Central-zone internal mass

Object	Construction	Entered surface area [m²]
HIBO Same-Zone Partition 104	HIBO Internal Mass Half SA6	14.920228
HIBO Same-Zone Door 104	HIBO Internal Mass Half HM Door	3.901928
HIBO Same-Zone Partition 102	HIBO Internal Mass Half SA6	11.650041
HIBO Same-Zone Door 102	HIBO Internal Mass Half HM Door	3.901928

4. Envelope Materials and Constructions

Opaque heat-transfer surfaces use layer-by-layer Construction definitions. The exterior surface film is not inserted as a material layer; the model selects TARP for both inside and outside surface convection. FullExterior solar distribution and the shading settings in Section 2.1 are used. Equivalent materials remain explicit model representations rather than implied product tests. [1]

4.1 Exterior Walls

The metal-panel exterior wall has six layers from exterior to interior: equivalent sheet-metal skin, continuous insulation, an air-barrier resistance, exterior gypsum sheathing, an effective steel-stud/batt cavity, and interior Type X gypsum. The CMU-veneer wall has seven layers, with equivalent hollow CMU and a drainage cavity outside the same insulated/sheathing/cavity/gypsum sequence. The assemblies follow the drawing descriptions while the thermal tuples and equivalents are those entered in the IDF. [1,3]

4.2 Roofs

The roof uses EPDM membrane, four-inch R-30-equivalent polyisocyanurate, a resistance-only vapor-retarder layer, and equivalent metal decking. The modeled roof is sloped using surface vertices; the roof insulation layer remains 0.1016 m in the material definition. Material thickness is therefore not inferred from the elevation difference. [1,3]

4.3 Slab-on-Grade Floors

Both slab constructions start at the ground side with 0.1016 m granular fill followed by 0.1016 m concrete. The carpeted slab adds a 0.005207 m carpet-tile equivalent on the room side. Sealed-concrete floors do not add a separate massive sealer layer. All slab exterior boundaries are Ground, using the twelve monthly temperatures below; no Foundation:Kiva object is present. [1,3]

Table 4-1. Prescribed ground-contact temperatures

Month	Ground temperature [°C]
January	21.5
February	21.4
March	21.5
April	21.5
May	22
June	22.9
July	23
August	23.1
September	23.1
October	22.2
November	21.7
December	21.6

These temperatures belong to Site:GroundTemperature:BuildingSurface in the IDF. The separate ground-temperature information at several depths in the EPW header is not substituted for them. The source records do not establish these monthly boundary values as field measurements. [1,2]

4.4 Interior Partitions, Ceilings, and Construction Totals

Interior partitions use gypsum layers around six-inch or eight-inch effective sound-batt cavities. The suspended acoustic ceiling is represented by an Armstrong ULTIMA High NRC 1940 reference material; this reference item is not asserted to be the installed SKU. A separate gypsum ceiling construction is also used. Doors, frame strips, spandrels and same-zone half constructions have their own records. [1,3]

Table 4-2. All 17 construction definitions

Construction	Layers: exterior / adjacent side → room side
EXT_Window_SN54	SN54_1in_IGU
HIBO Metal Panel Exterior Wall	Metal Wall Panel 24ga Eq → Thermax CI R7.5 Eq → Air Barrier Eq → Exterior Gypsum Sheathing 5-8in → CFS Batt R13 6in Effective → Gypsum Board Type X 5-8in
HIBO CMU Veneer Exterior Wall	Burnished CMU Veneer 4in Eq → Exterior Drainage Cavity → Thermax CI R7.5 Eq → Air Barrier Eq → Exterior Gypsum Sheathing 5-8in → CFS Batt R13 6in Effective → Gypsum Board Type X 5-8in
HIBO EPDM Insulated Roof	EPDM Membrane 45mil → Roof Polyiso 4in R30 Eq → Roof Vapor Retarder Eq → Metal Roof Deck Eq
HIBO Slab Carpet	Granular Fill 4in Eq → Concrete Slab 4in → Carpet Tile CPT-1 Eq
HIBO Slab Sealed Concrete	Granular Fill 4in Eq → Concrete Slab 4in
HIBO APC-1 Ceiling	APC-1 Ultima High NRC 1940 Reference
HIBO GYP Ceiling	Gypsum Board Type X 5-8in
HIBO Interior Partition SA.6.20	Gypsum Board Type X 5-8in → Interior Sound Batt Cavity 6in Eq → Gypsum Board Type X 5-8in
HIBO Interior Partition SA.8.21	Gypsum Board Type X 5-8in → Interior Sound Batt Cavity 8in Eq → Gypsum Board Type X 5-8in
HIBO Interior CTG	Interior CTG 1-4in
HIBO HM Door 1-3-4in	HM Door Steel Skin Eq → HM Door Core Eq → HM Door Steel Skin Eq
HIBO Metal Frame Strip	Hollow Metal Frame Eq
HIBO Exterior Aluminum Frame Strip	Exterior Aluminum Frame Strip Eq
HIBO Internal Mass Half SA6	Internal Mass Half Cavity SA6 Eq → Gypsum Board Type X 5-8in
HIBO Internal Mass Half HM Door	Internal Mass Half HM Core → HM Door Steel Skin Eq
HIBO ITSG Opaque Panel	ITSG Opaque Panel 1in Eq

Table 4-3. Calculated construction resistance and areal heat capacity

Opaque construction	R, no films [m² K/W]	C_A [kJ/(m² K)]
HIBO Metal Panel Exterior Wall	2.576922	36.207
HIBO CMU Veneer Exterior Wall	2.917107	101.279
HIBO EPDM Insulated Roof	5.288905	9.126
HIBO Slab Carpet	0.217755	338.816
HIBO Slab Sealed Concrete	0.130972	336.746
HIBO APC-1 Ceiling	0.390000	3.284
HIBO GYP Ceiling	0.099219	13.843
HIBO Interior Partition SA.6.20	1.468437	31.953
HIBO Interior Partition SA.8.21	1.891771	33.376
HIBO HM Door 1-3-4in	0.353794	11.220
HIBO Metal Frame Strip	0.200000	5.556
HIBO Exterior Aluminum Frame Strip	0.200000	16.459

Opaque construction	R, no films [m ² K/W]	C_A [kJ/(m ² K)]
HIBO Internal Mass Half SA6	0.734219	15.977
HIBO Internal Mass Half HM Door	0.176897	5.610
HIBO ITSG Opaque Panel	0.418235	8.534

Source: [1, L1296–L1387]. Calculations use $R = \Sigma(t/k)$ plus resistance-only layers and $C_A = \Sigma(tpc_p)/1000$. NoMass and AirGap layers add their entered resistance but no explicit material heat capacity to this calculation. Glazing is not treated as an opaque massive layer. For slabs, the listed heat capacity includes the modeled granular-fill layer. These construction totals are not a single whole-building equivalent R or C.

4.5 Glazing, Frames, Doors, and Spandrel

The exterior SN54 simple-glazing record specifies glazing performance; it is not the frame-inclusive whole-window result. Interior clear glass uses a detailed spectral-average optical record. The main CWB/CWC windows use FrameAndDivider objects, while many other frames are explicit opaque strips. Insulated spandrels are opaque panels, not solar-transmitting glass. [1]

Table 4-4. Glazing thermal and optical inputs

Glazing record	Property	Input	Source / basis
SN54_1in_IGU	U-factor [W/(m ² K)]	1.36	Guardian SN54 on clear, double-glazed reference
SN54_1in_IGU	SHGC / visible transmittance	0.28 / 0.54	Guardian SN54 reference
HIBO Interior CTG 1-4in	Thickness [mm]	6.35	A0.1 §088000: 1/4 in clear tempered glass
HIBO Interior CTG 1-4in	Conductivity [W/(mK)]	0.9	EnergyPlus 22.2 WindowGlassMaterials: CLEAR 6MM reference
HIBO Interior CTG 1-4in	Solar transmittance / front and back reflectance	0.775 / 0.071 / 0.071	CLEAR 6MM reference
HIBO Interior CTG 1-4in	Visible transmittance / front and back reflectance	0.881 / 0.08 / 0.08	CLEAR 6MM reference
HIBO Interior CTG 1-4in	IR transmittance / front and back emissivity	0 / 0.84 / 0.84	CLEAR 6MM reference
HIBO Interior CTG 1-4in	Optical data / dirt factor / solar diffusing	SpectralAverage / 1 / No	IDF settings

Table 4-5. Main window frame and divider inputs

Input	CWB Vision Frame	CWC Vision Frame
Frame width	0.0508 m	0.0508 m
Frame conductance	5.0 W/(m ² K)	5.0 W/(m ² K)
Frame inside/outside projection	0.0 / 0.0 m	0.0 / 0.0 m
Frame edge-to-center conductance ratio	1.3	1.3
Frame solar / visible absorptance / emissivity	0.40 / 0.40 / 0.90	0.40 / 0.40 / 0.90
Divider type / width	DividedLite / 0.0508 m	DividedLite / 0.0508 m
Horizontal / vertical divider count	2 / 0	2 / 3
Divider inside/outside projection	0.0 / 0.0 m	0.0 / 0.0 m
Divider conductance / edge ratio	5.0 W/(m ² K) / 1.3	5.0 W/(m ² K) / 1.3
Divider solar / visible absorptance / emissivity	0.40 / 0.40 / 0.90	0.40 / 0.40 / 0.90
Outside reveal solar absorptance	0.50	0.50

Input	CWB Vision Frame	CWC Vision Frame
Inside sill depth / solar absorptance	0.0 m / 0.50	0.0 m / 0.50
Inside reveal depth / solar absorptance	0.0 m / 0.50	0.0 m / 0.50

Sources: [1, L1266–L1291, L5681–L5735]. The 5 W/(m² K) frame/divider conductance, 1.3 edge ratios, and explicit strip equivalents are model inputs, not certified installed assembly performance. Opaque frame strips use their own assumed layer properties, while the detailed exterior simple-glazing model and internal glass are independent material records.

4.6 Material Properties and Sources

Tables 4-6 through 4-27 list material thickness, conductivity, density, specific heat, and surface properties with the source recorded for each value. Equivalent layers and generic reference properties are identified in the corresponding rows; they are not installed-product certifications. [1,3]

Table 4-6. HIBO Metal Wall Panel 24ga Eq

Property	Input	Unit	Source / basis
Thickness	0.61	mm	Equivalent skin thickness; not independently verified gauge
Conductivity	45.28	W/(m K)	ASHRAE HOF F08 generic steel
Density	7824	kg/m ³	ASHRAE HOF F08 generic steel
Specific heat	500	J/(kg K)	ASHRAE HOF F08 generic steel
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L685–L694].

Table 4-7. HIBO Burnished CMU Veneer 4in Eq

Property	Input	Unit	Source / basis
Thickness	92.1	mm	4 in hollow-wall equivalent
Conductivity	0.484229679367	W/(m K)	CMHA TEK 06-02C, derived t/R without films
Density	832.318730782	kg/m ³	CMHA TEK 06-16A heat capacity, derived bulk-effective density
Specific heat	880	J/(kg K)	CMHA TEK 06-16A masonry reference
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L718–L727].

Table 4-8. HIBO Thermax CI R7.5 Eq

Property	Input	Unit	Source / basis
Thickness	27.2	mm	A0.1 R-7.5 equivalent / DuPont THERMAX R-7 per inch
Conductivity	0.0205931898624	W/(m K)	Derived from A0.1 R-7.5 and modeled thickness
Density	32	kg/m ³	ASHRAE HOF faced-polyiso generic reference
Specific heat	920	J/(kg K)	ASHRAE HOF faced-polyiso generic reference
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L745–L754].

Table 4-9. HIBO Exterior Gypsum Sheathing 5-8in

Property	Input	Unit	Source / basis
Thickness	15.875	mm	A1.1/A1.2: 5/8 in
Conductivity	0.16	W/(m K)	ASHRAE HOF G01/G01a generic gypsum
Density	800	kg/m ³	ASHRAE HOF G01/G01a generic gypsum
Specific heat	1090	J/(kg K)	ASHRAE HOF G01/G01a generic gypsum
Roughness	MediumSmooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.4 / 0.4	—	IDF surface settings

[1, L774–L783].

Table 4-10. HIBO CFS Batt R13 6in Effective

Property	Input	Unit	Source / basis
Thickness	152.4	mm	A1.2: 6 in framing
Conductivity	0.14423	W/(m K)	Assumed effective steel/batt cavity resistance ≈ R-6 (IP)
Density	50	kg/m ³	Assumed effective composite density
Specific heat	700	J/(kg K)	Assumed effective composite specific heat
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.5 / 0.5	—	IDF surface settings

[1, L795–L804].

Table 4-11. HIBO Gypsum Board Type X 5-8in

Property	Input	Unit	Source / basis
Thickness	15.875	mm	A1.1/A1.2: 5/8 in Type X
Conductivity	0.16	W/(m K)	ASHRAE HOF G01/G01a generic gypsum
Density	800	kg/m ³	ASHRAE HOF G01/G01a generic gypsum
Specific heat	1090	J/(kg K)	ASHRAE HOF G01/G01a generic gypsum
Roughness	MediumSmooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.4 / 0.4	—	IDF surface settings

[1, L824–L833].

Table 4-12. HIBO EPDM Membrane 45mil

Property	Input	Unit	Source / basis
Thickness	1.143	mm	A0.1 §075323: 45 mil EPDM
Conductivity	0.25	W/(m K)	EN 12524:2000, Table 1 EPDM reference
Density	1150	kg/m ³	EN 12524:2000, Table 1 EPDM reference
Specific heat	1000	J/(kg K)	EN 12524:2000, Table 1 EPDM reference
Roughness	VeryRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L852–L861].

Table 4-13. HIBO Roof Polyiso 4in R30 Eq

Property	Input	Unit	Source / basis
Thickness	101.6	mm	A1.2: 4 in roof insulation
Conductivity	0.0192304052391	W/(m K)	Derived to preserve A0.1 R-30 target
Density	32	kg/m ³	ASHRAE HOF faced-polyiso generic reference
Specific heat	920	J/(kg K)	ASHRAE HOF faced-polyiso generic reference
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L878–L887].

Table 4-14. HIBO Metal Roof Deck Eq

Property	Input	Unit	Source / basis
Thickness	1.5	mm	EnergyPlus 22.2 DOE reference: Metal Decking
Conductivity	45.006	W/(m K)	EnergyPlus 22.2 DOE reference: Metal Decking
Density	7680	kg/m ³	EnergyPlus 22.2 DOE reference: Metal Decking
Specific heat	418.4	J/(kg K)	EnergyPlus 22.2 DOE reference: Metal Decking
Roughness	MediumSmooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.3	—	IDF surface settings

[1, L908–L917].

Table 4-15. HIBO Concrete Slab 4in

Property	Input	Unit	Source / basis
Thickness	101.6	mm	A1.2: 4 in concrete slab
Conductivity	1.311	W/(m K)	EnergyPlus 22.2 DOE reference: MAT-CC05
Density	2240	kg/m ³	EnergyPlus 22.2 DOE reference: MAT-CC05
Specific heat	836.8	J/(kg K)	EnergyPlus 22.2 DOE reference: MAT-CC05
Roughness	Rough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L937–L946].

Table 4-16. HIBO Granular Fill 4in Eq

Property	Input	Unit	Source / basis
Thickness	101.6	mm	A1.2: 4 in granular fill
Conductivity	1.9	W/(m K)	EnergyPlus 22.2 Basement material reference: gravel
Density	2000	kg/m ³	EnergyPlus 22.2 Basement material reference: gravel
Specific heat	720	J/(kg K)	EnergyPlus 22.2 Basement material reference: gravel
Roughness	Rough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L965–L974].

Table 4-17. HIBO APC-1 Ultima High NRC 1940 Reference

Property	Input	Unit	Source / basis
Thickness	22.225	mm	Armstrong ULTIMA High NRC 1940 reference: 7/8 in
Conductivity	0.0569871794872	W/(m K)	Derived t/R from manufacturer reference R = 0.39 m ² K/W
Density	250.437233092	kg/m ³	Derived from manufacturer reference areal weight
Specific heat	590	J/(kg K)	ASHRAE HOF F16 generic acoustic tile
Roughness	MediumSmooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.3 / 0.13	—	IDF surface settings

[1, L994–L1003].

Table 4-18. HIBO Interior Sound Batt Cavity 6in Eq

Property	Input	Unit	Source / basis
Thickness	152.4	mm	A1.1: SA.6.20 framing tag
Conductivity	0.12	W/(m K)	Assumed effective cavity
Density	40	kg/m ³	Assumed composite density
Specific heat	700	J/(kg K)	Assumed composite specific heat
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.5 / 0.5	—	IDF surface settings

[1, L1015–L1024].

Table 4-19. HIBO Interior Sound Batt Cavity 8in Eq

Property	Input	Unit	Source / basis
Thickness	203.2	mm	A1.1: SA.8.21 tag interpretation; schedule thickness conflicts
Conductivity	0.12	W/(m K)	Assumed effective cavity
Density	40	kg/m ³	Assumed composite density
Specific heat	700	J/(kg K)	Assumed composite specific heat
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.5 / 0.5	—	IDF surface settings

[1, L1036–L1045].

Table 4-20. HIBO HM Door Steel Skin Eq

Property	Input	Unit	Source / basis
Thickness	1	mm	Assumed 1 mm steel skin
Conductivity	45.28	W/(m K)	ASHRAE HOF F08 generic steel
Density	7824	kg/m ³	ASHRAE HOF F08 generic steel
Specific heat	500	J/(kg K)	ASHRAE HOF F08 generic steel
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L1065–L1074].

Table 4-21. HIBO HM Door Core Eq

Property	Input	Unit	Source / basis
Thickness	42.45	mm	44.45 mm leaf minus two assumed 1 mm skins
Conductivity	0.12	W/(m K)	Assumed unknown door core
Density	80	kg/m ³	Assumed unknown door core
Specific heat	1000	J/(kg K)	Assumed unknown door core
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.5 / 0.5	—	IDF surface settings

[1, L1086–L1095].

Table 4-22. HIBO Hollow Metal Frame Eq

Property	Input	Unit	Source / basis
Thickness	44.45	mm	Assumed equivalent frame depth
Conductivity	0.22225	W/(m K)	Derived from assumed layer R = 0.20 m ² K/W
Density	250	kg/m ³	Assumed hollow-section bulk density
Specific heat	500	J/(kg K)	Generic steel specific-heat proxy
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.4 / 0.4	—	IDF surface settings

[1, L1108–L1117].

Table 4-23. HIBO Exterior Aluminum Frame Strip Eq

Property	Input	Unit	Source / basis
Thickness	152.4	mm	A0.1: nominal 6 in storefront-equivalent depth
Conductivity	0.762	W/(m K)	Derived from assumed layer R = 0.20 m ² K/W
Density	120	kg/m ³	Assumed hollow-section bulk density
Specific heat	900	J/(kg K)	NIST WebBook aluminum at 25°C; rounded
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.4 / 0.4	—	IDF surface settings

[1, L1130–L1139].

Table 4-24. HIBO Internal Mass Half Cavity SA6 Eq

Property	Input	Unit	Source / basis
Thickness	76.2	mm	Half SA6 cavity
Conductivity	0.12	W/(m K)	Same assumed SA6 cavity
Density	40	kg/m ³	Same assumed SA6 cavity
Specific heat	700	J/(kg K)	Same assumed SA6 cavity
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.5 / 0.5	—	IDF surface settings

[1, L1151–L1160].

Table 4-25. HIBO Internal Mass Half HM Core

Property	Input	Unit	Source / basis
Thickness	21.225	mm	Half HM core
Conductivity	0.12	W/(m K)	Same assumed HM core
Density	80	kg/m ³	Same assumed HM core
Specific heat	1000	J/(kg K)	Same assumed HM core
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.5 / 0.5	—	IDF surface settings

[1, L1172–L1181].

Table 4-26. HIBO ITSG Opaque Panel 1in Eq

Property	Input	Unit	Source / basis
Thickness	25.4	mm	A0.1 \$088000: 1 in opaque spandrel
Conductivity	0.0607313642757	W/(m K)	Derived from assumed $U = 1.70 \text{ W}/(\text{m}^2\text{K})$ and film $R = 0.17 \text{ m}^2\text{K}/\text{W}$
Density	400	kg/m ³	Assumed spandrel composite
Specific heat	840	J/(kg K)	Assumed spandrel composite
Roughness	Smooth	—	IDF surface setting
Absorptance: thermal / solar / visible	0.84 / 0.6 / 0.6	—	IDF surface settings

[1, L1193–L1202].

Table 4-27. HIBO Carpet Tile CPT-1 Eq

Property	Input	Unit	Source / basis
Thickness	5.207	mm	J+J Kinetex 1840 reference: nominal 0.205 in
Conductivity	0.06	W/(m K)	ASHRAE HOF F17 generic carpet
Density	288	kg/m ³	ASHRAE HOF F17 generic carpet
Specific heat	1380	J/(kg K)	ASHRAE HOF F17 generic carpet
Roughness	MediumRough	—	IDF surface setting
Absorptance: thermal / solar / visible	0.9 / 0.7 / 0.7	—	IDF surface settings

[1, L1222–L1231].

The CFS/batt material's name contains R13, but its entered conductivity represents an effective framed cavity rather than unbridged R-13 insulation. The CMU density is a bulk-effective hollow-assembly value. Interior batt cavities, door cores, frame strips, and spandrel composites retain their declared equivalent-property assumptions. These choices are not silently replaced by nominal material labels. [1]

Table 4-28. Resistance-only material inputs

Layer	R [m²K/W]	Roughness	Thermal / solar / visible absorptance	Source / basis
HIBO Air Barrier Eq	0.001	MediumSmooth	0.90 / 0.50 / 0.50	Thin-membrane modeling assumption; [1, L1237–L1243]
HIBO Roof Vapor Retarder Eq	0.0010	Smooth	0.90 / 0.50 / 0.50	Thin-membrane modeling assumption; [1, L1246–L1252]
HIBO Exterior Drainage Cavity	0.15	Not applicable	Not applicable	EnergyPlus 22.2 ASHRAE HOF F04 wall air space; [1, L1259–L1261]

5. Occupancy, Lighting, and Equipment

5.1 Occupancy

Each of the three main zones has one People object, using the same area-per-person method, people multiplier, and activity schedule. Neither MEP nor the plenum has a People object. The occupancy inputs are schedule assumptions in the model, not measured occupant counts. [1]

Table 5-1. People inputs common to the three main zones

Parameter	Input	Source / basis
Calculation method	Area/Person	[1, L5859–L5902]
Floor area per person	18.580626 m ² /person	IDF occupancy-density input
Number-of-people schedule	OCCUPY-1	Schedule multiplier applied to area-based design count
Activity schedule	ActSchd	Activity level defined in Appendix A
Radiant fraction	0.30	IDF sensible-heat split input
Sensible heat fraction	Blank	Not assigned a constant value in the IDF
Mean radiant temperature calculation	ZoneAveraged	IDF setting
ASHRAE 55 comfort warnings	No	IDF setting

Table 5-2. Calculated design occupancy from area and activity inputs

Zone	Design people count	Full-schedule activity heat [W]
Thermal Zone: Space 103 LAB B	0.991	116.19
Thermal Zone: Space 102 - Plus	1.318	154.57
Thermal Zone: Space 105 LAB A	0.991	116.19

The calculated count is fractional because an area-per-person density is used. The activity heat is total metabolic heat before the solver determines the sensible/latent split; it is not all convective sensible gain. ActSchd is 117.239998 W/person all day. The sensible-heat-fraction field is blank, and the CO₂ generation-rate field is also blank. [1]

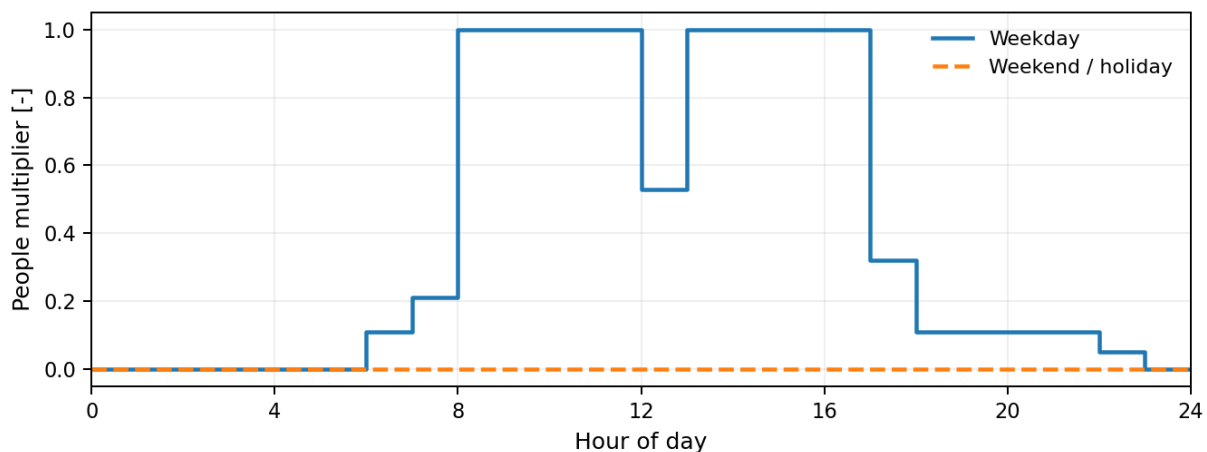


Figure 5-1. OCCUPY-1 weekday and weekend profiles. Exact day-type and design-day values are in Appendix A.1. [1]

5.2 Electric Lighting

The three main zones each contain a Lights object using Watts/Area. The lighting density is multiplied by `ltg_sch_office`. No `Daylighting:Controls` object is instantiated, so the daylight availability at the windows does not imply an active daylight-dimming control. The plenum and MEP have no Lights objects. [1]

Table 5-3. Interior lighting inputs

Parameter	Input	Source / basis
Design-level method	Watts/Area	[1, L5907–L5950]
Lighting power density	6.181254 W/m ²	IDF load input
Schedule	<code>ltg_sch_office</code>	Appendix A
Return-air / radiant / visible fractions	0 / 0.70 / 0.20	IDF fractions; remainder 0.10 is convective
Fraction replaceable	1	IDF setting
End-use subcategory	LightsWired	IDF setting

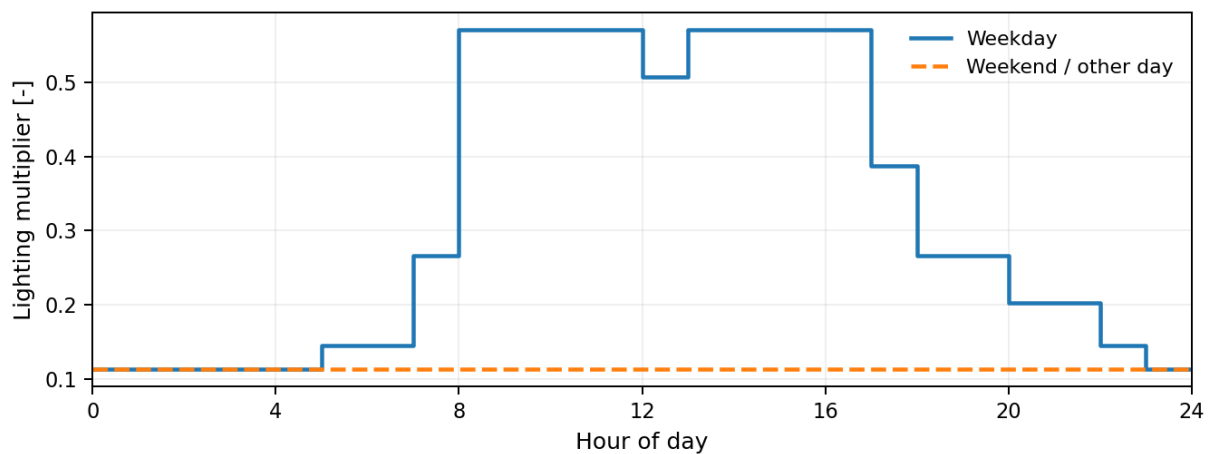


Figure 5-2. Interior lighting schedule. The entered weekday maximum is about 0.571 of design power, not 1.0. [1]

Table 5-4. Calculated zone design powers before schedule multipliers

Zone	Calculated design lighting [W]	Calculated design plug load [W]
Thermal Zone: Space 103 LAB B	113.82	124.85
Thermal Zone: Space 102 - Plus	151.42	166.09
Thermal Zone: Space 105 LAB A	113.82	124.85

`Exterior_Lights_a` is a separate 88.49 W load. Its schedule is zero from 00:00–06:00 and one from 06:00–24:00 for `AllDays`. The `Control Option` field is blank. The exterior load is not applied as a zone internal heat gain. [1, L498–L506, L6056–L6061]

5.3 Plug and Process Loads

One `ElectricEquipment` object is assigned to each main zone. Each uses a 6.78 W/m² design power and the `EQUIP-1` schedule. The source does not enumerate individual laboratory devices or provide a measured plug-load inventory. Latent, radiant and lost fraction fields are blank, so the report does not substitute a measured heat split. [1]

Table 5-5. Electric-equipment inputs

Parameter	Input	Source / basis
Design-level method	Watts/Area	[1, L5955–L5992]
Equipment power density	6.78 W/m ²	IDF load input
Schedule	EQUIP-1	Appendix A
Latent / radiant / lost fractions	Blank / Blank / Blank	Fields are not explicitly entered; no measured heat split is asserted
End-use subcategory	MiscPlug	IDF setting

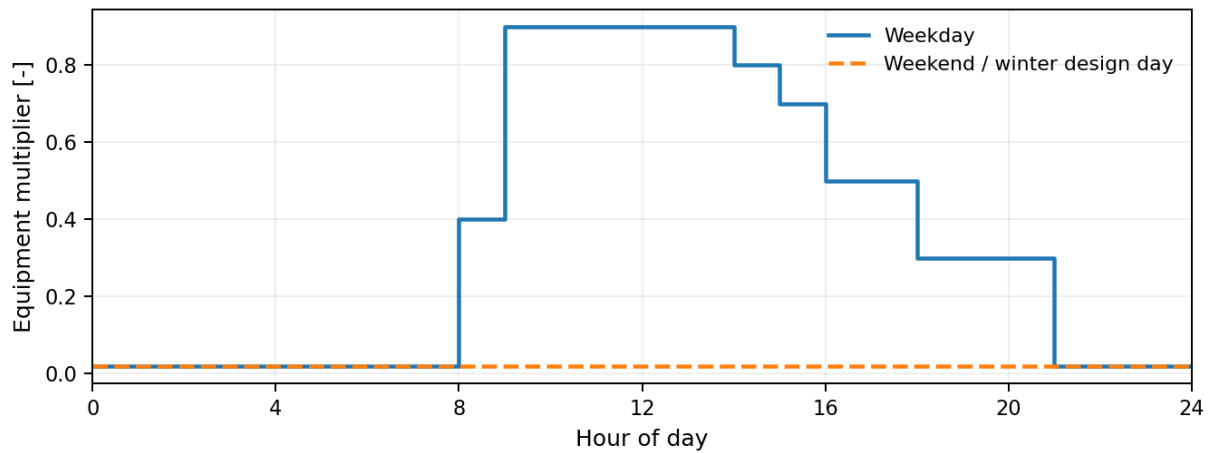


Figure 5-3. EQUIP-1 operating profile. Weekday, summer design-day and custom-day profiles share the plotted daytime curve. [1]

6. HVAC, Controls, and Air Exchange

6.1 System Arrangement

The air loop DXVAV Sys 1 serves all three main zones. Zone return nodes enter a common return plenum; air passes through a series return fan, outdoor/return-air mixer, DX cooling coil, central electric heater, supply fan, and zone splitter. Each branch then passes through a VAV damper and electric reheat coil before entering its zone. [1]

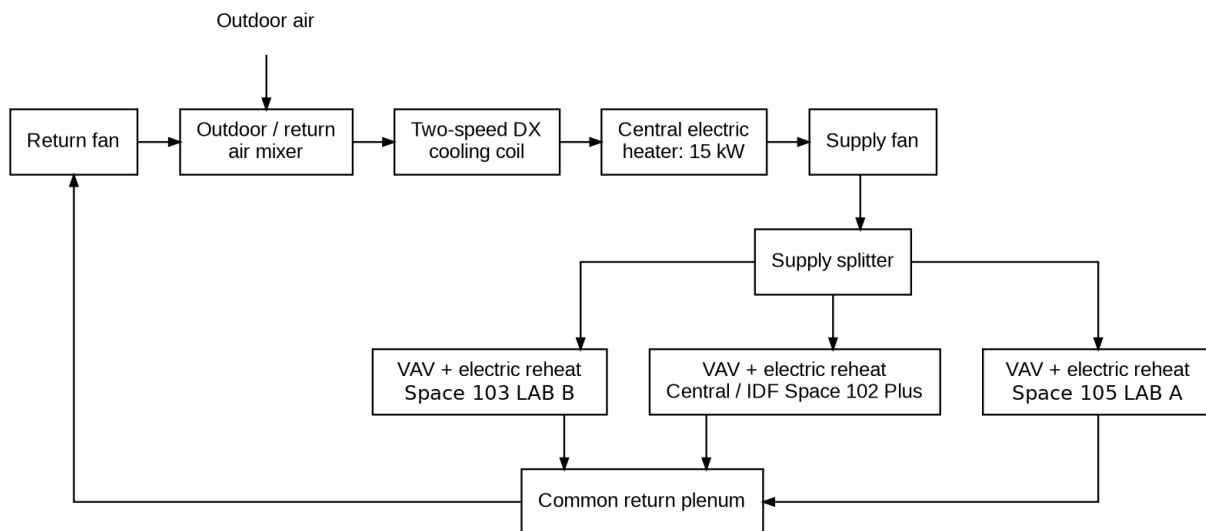


Figure 6-1. Implemented common air loop and three VAV/reheat branches. [1]

Table 6-1. Terminal-to-zone mapping

Zone	Terminal name	Terminal reheat coil
Thermal Zone: Space 103 LAB B	Thermal Zone: Space 103 LAB B VAV Reheat	Thermal Zone: Space 103 LAB B Reheat Coil
Thermal Zone: Space 102 - Plus	Thermal Zone: Space 102 - Plus VAV Reheat	Thermal Zone: Space 102 - Plus Reheat Coil
Thermal Zone: Space 105 LAB A	Thermal Zone: Space 105 LAB A VAV Reheat	Thermal Zone: Space 105 LAB A Reheat Coil

SupplyPath, ReturnPath, Branch, BranchList, NodeList and ZoneHVAC equipment-connection objects implement this routing. Appendix B.1 gives the common node sequence and points to the full connection definitions. MEP has no dedicated supply terminal. [1]

6.2 Cooling and Electric Heating

Table 6-2. Cooling and heating component inputs

Component / parameter	Input	Source / basis
DX coil object	Coil:Cooling:DX:TwoSpeed / DXVAV Sys 1 Cooling Coil	[1, L6497–L6530]
High / low gross cooling capacity	Autosize / Autosize	Calculated by EnergyPlus; no fixed nameplate tonnage is entered
High / low rated airflow	Autosize / Autosize	IDF sizing inputs
High / low sensible heat ratio	Autosize / Autosize	IDF sizing inputs
High / low gross COP	3.28 / 3.28 W/W	IDF performance inputs

Component / parameter	Input	Source / basis
Condenser type	AirCooled	IDF setting
Minimum compressor outdoor dry bulb	–25°C	IDF setting
DX coil availability schedule	Blank	No schedule name is entered in the coil object
Central electric heating coil	DXVAV Sys 1 Heating Coil	Coil:Heating:Electric
Central heating nominal capacity	15,000 W	AAON RTU-1 nameplate: maximum electric heat 15 kW
Central heating efficiency / schedule	1.0 / AvailSched_1	IDF input
Terminal electric reheaters	One coil per main zone; nominal capacity = Autosize	[1, L6535–L6566]
Terminal reheat efficiency / schedule	1.0 / AvailSched_1	IDF input

The DX coil uses gross COP 3.28 at both speeds and AirCooled condenser type. Internal static pressure, condenser-air-inlet name, evaporative-condenser fields, water tank references and basin-heater capacity are blank; the basin-heater setpoint field is 2°C. The associated CoilSystem:Cooling:DX has an empty availability-schedule name and connects the mixed-air and cooling-coil-outlet nodes. Its sensor node is the cooling-coil outlet; Dehumidification Control Type is None, Run on Sensible Load is Yes, and Run on Latent Load is No. Additional explicitly entered and blank fields are preserved in the complete input register. [1, L6497–L6530, L6571–L6581]

Table 6-3. Equipment-source evidence versus implemented capacity inputs

Equipment evidence	Documented value	How it is used in this IDF
Installed AAON RTU electric heat	15 kW maximum output	Central nominal heating capacity = 15,000 W
Installed RTU field note	Reconfigured from 6 ton to 4 ton	Not imposed as a fixed DX cooling capacity; DX capacity remains Autosize
Nameplate maximum outlet-air temperature	200°F	Rating on the central unit; not entered as a terminal DAT control limit
M1.1 terminal schedule TB-1 / TB-2 / TB-3	8.9 / 4.4 / 8.9 kW; 800 / 400 / 800 CFM maximum	Design evidence, not the current autosized terminal inputs
M1.1 RTU basis of design	Daikin DPS005A	Different evidence from the installed AAON label

Sources: [1,3,4]. The central heater and the three terminal reheaters are separate components. A nameplate maximum is not continuous electric consumption. The source schedule's approximately 90°F terminal leaving-air condition is not asserted as an active discharge-temperature ceiling.

6.3 Supply and Return Fans and Availability

Table 6-4. Supply and return fan inputs

Parameter	Supply fan	Return fan
Object type	Fan:VariableVolume	Fan:VariableVolume
Maximum flow rate	Autosize	Autosize
Pressure rise	1,389.42 Pa	1,389.42 Pa
Total efficiency	0.43	0.30
Motor efficiency	0.815	0.85
Motor-in-airstream fraction	1	1
Power minimum-flow method / fraction	Fraction / 0.40	Fraction / 0.40
Availability schedule	AvailSched_1	AvailSched_1
Power coefficients c1...c5	0.0408, 0.088, –0.0729, 0.9437, 0	Same

The fan polynomial uses the listed coefficients in the entered order. The 0.40 minimum-flow fraction belongs to the fan-power model and is separate from the terminal minimum-airflow fraction. Both fan availability schedules are

AvailSched_1, whose profiles are one throughout the year. The model therefore does not implement the drawing's normal unoccupied fan-off sequence through this schedule. [1,3]

Table 6-5. Availability-manager inputs

Availability-manager input	Value
Object	DXVAV Sys 1 Availability
Applicability schedule	HVACTemplate-Always 1
Fan schedule	AvailSched_1
Night-cycle control	StayOff
Thermostat tolerance	1°C
Cycling control / run time	FixedRunTime / 1,800 s

Source: [1, L6456–L6492, L6773–L6788]. The NightCycle manager's Control Type is StayOff; it is not described as an optimum-start controller. The nominal cycling tolerance and time are retained as input fields, but no additional night-cycle behavior is inferred from their presence alone.

No dedicated bathroom-exhaust Fan:ZoneExhaust object is connected. The M1.1 design schedule lists EF-1 at 125 CFM and 42.4 W; that device is not a separate load in this model. The series return fan is not an established equivalent of the installed RTU power-exhaust motor. [1,3,4]

6.4 VAV Terminals

Table 6-6. VAV terminal inputs common to all three zones

Parameter	Input
Maximum airflow	Autosize
Minimum-airflow input method	Constant
Constant minimum-airflow fraction	Autosize
Fixed minimum flow / minimum-fraction schedule	Blank / Blank
Terminal availability schedule	Blank
Reheat coil type	Coil:Heating:Electric
Damper Heating Action	Reverse
Maximum flow per zone floor area during reheat	Autocalculate
Maximum flow fraction during reheat	Blank
Maximum Reheat Air Temperature	Blank; no value entered
Convergence tolerance	0.001
Outdoor-air object	Matching SZ DSOA for each zone

Source: [1, L6306–L6370]. The entered maximum reheat-air-temperature field is blank. The 40°C heating design supply temperature in Sizing:Zone is a sizing condition, not a terminal operating ceiling. Damper Heating Action is reported literally as Reverse; it is not used here to assert an independently verified BAS sequence for the electric reheaters.

6.5 Supply-Air Temperature Control

Table 6-7. Temperature-control inputs

Control input	Value	Source
AHU setpoint manager	SetpointManager:Warmest / DXVAV Sys 1 SAT Reset Manager	[1, L6812–L6819]
Minimum / maximum SAT target	12.0 / 17.7°C	IDF setpoint bounds
Reset strategy	MaximumTemperature	IDF strategy
Fan-heat compensation	SetpointManager:MixedAir objects assign upstream coil-node targets	[1, L6793–L6807]
Zone thermostat	ThermostatSetpoint:DualSetpoint / Dual SP Control	[1, L6298–L6301]
Heating / cooling schedule	HTGSETP_SCH_NO_OPTIMUM / CLGSETP_SCH_NO_OPTIMUM	Schedule values are given in Appendix A
Thermostat control schedule	HVACTemplate-Always 4	[1, L6274–L6293]

The warmest-zone manager resets the AHU supply-air target within 12.0–17.7°C. The two MixedAir managers connect that reference to upstream coil nodes with the supply-fan inlet and outlet specified for compensation. Zone thermostat schedules are given once in Section 6.6 and Appendix A.2. [1]

6.6 Zone Thermostat Set Points

All three main zones use ThermostatSetpoint:DualSetpoint through the constant control selector HVACTemplate-Always 4. The heating and cooling schedules have different transition times: weekday heating remains at 21.11°C until 19:00, while weekday cooling returns to 29.44°C at 18:00. These are implemented schedules, not a statement of the present BAS operating program. [1]

Table 6-8. Thermostat schedule summary

Control	Schedule	Weekday daytime	Other periods
Heating	HTGSETP_SCH_NO_OPTIMUM	21.11°C, 06:00–19:00	15.56°C
Cooling	CLGSETP_SCH_NO_OPTIMUM	23.89°C, 06:00–18:00	29.44°C
Control type	HVACTemplate-Always 4	4, all day	4, all day

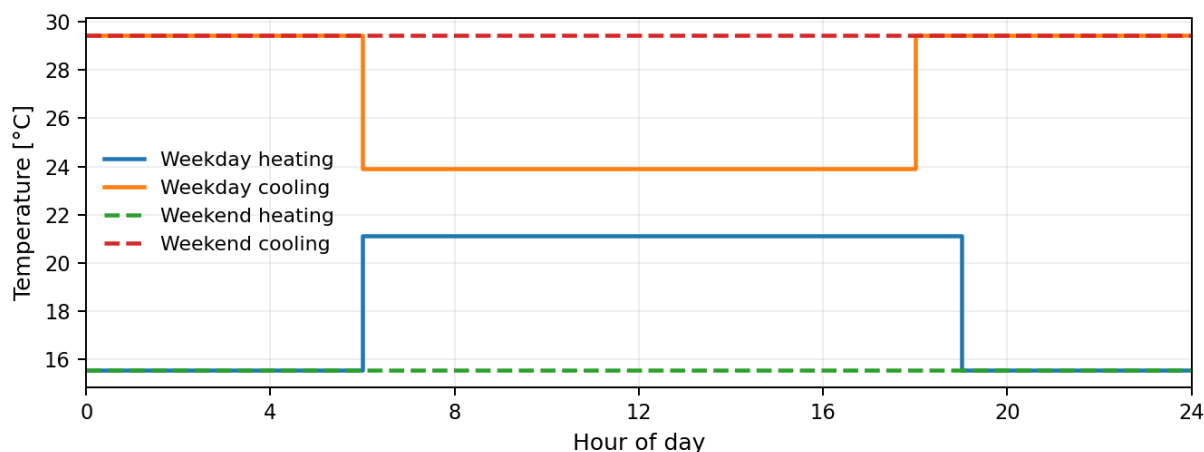


Figure 6-2. Zone thermostat schedules. Design-day and custom-day cases are tabulated in Appendix A.2. [1]

There is no separate humidity setpoint or humidification controller. The common AHU supply-air reset is a different control from the zone thermostat and is described in Section 6.5. [1]

6.7 Ventilation and Total Airflow

Mechanical ventilation is requested through three DesignSpecification:OutdoorAir objects and the connected Controller:MechanicalVentilation. The per-person and per-area components use the Sum method. These outdoor-air requests are different from the total recirculating supply flow, which is autosized at the system and terminal levels. [1]

Table 6-9. Ventilation and total-flow inputs

Input	Value
Zone outdoor-air method	Sum
Outdoor air per person	0.0025 m ³ /(s·person)
Outdoor air per floor area	0.0003 m ³ /(s·m ²)
Fixed flow / ACH component	0 / 0
Outdoor-air rate schedule	Min OA Sched; Appendix A.5
Cooling / heating air-distribution effectiveness	1 / 1
Secondary recirculation fraction	Blank
System method	Standard62.1VentilationRateProcedure
Demand controlled ventilation	No
Zone maximum outdoor-air fraction	1.0
AirLoopHVAC design supply airflow	Autosize
Terminal maximum flow / constant minimum fraction	Autosize / Autosize

Sources: [1, L6611–L6625]; [1, L6066–L6109, L6306–L6370, L6638–L6648]. Min OA Sched is 1 from 06:00–23:00 on its weekday/custom-day profile and 0 on weekends/holidays; both design-day profiles are 1. This outdoor-air schedule does not turn the supply and return fans off. The zero fixed minimum in Controller:OutdoorAir is not a substitute for the connected zone ventilation requests.

6.8 Economizer

Table 6-10. Outdoor-air controller and economizer inputs

Parameter	Entered value
Controller	DXVAV Sys 1 OA Controller
Economizer type / action	FixedDryBulb / ModulateFlow
Dry-bulb lower / upper limits	4.0 / 12.7778°C
Lockout type	NoLockout
Minimum / maximum outdoor-air flow	0.0 m ³ /s / Autosize
Minimum limit type	FixedMinimum
Minimum / maximum fraction schedule	Blank / Blank
Enthalpy / dewpoint high limits	Blank / Blank
Time-of-day economizer schedule	Blank

Source: [1, L6586–L6606]. The outdoor-air system contains one mixer and uses the mechanical-ventilation controller described in Section 6.7. A refrigerant compressor minimum outdoor dry-bulb input of –25°C is part of the DX coil; it is not the economizer limit. [1]

6.9 Infiltration

Four ZoneInfiltration:DesignFlowRate objects serve Lab B, the central zone, Lab A, and MEP. No plenum infiltration object is present. Flow is specified per exterior wall area and modified by a schedule and the entered environmental coefficients, so it should not be described as a constant whole-zone ACH input. [1]

Table 6-11. Infiltration inputs

Parameter	Input
Method	Flow/ExteriorWallArea
Rate per exterior wall area	0.000199350288 m ³ /(s·m ²)
Schedule	INFIL-SCH
Constant / temperature / wind / wind ² coefficients	0 / 0 / 0.224 / 0

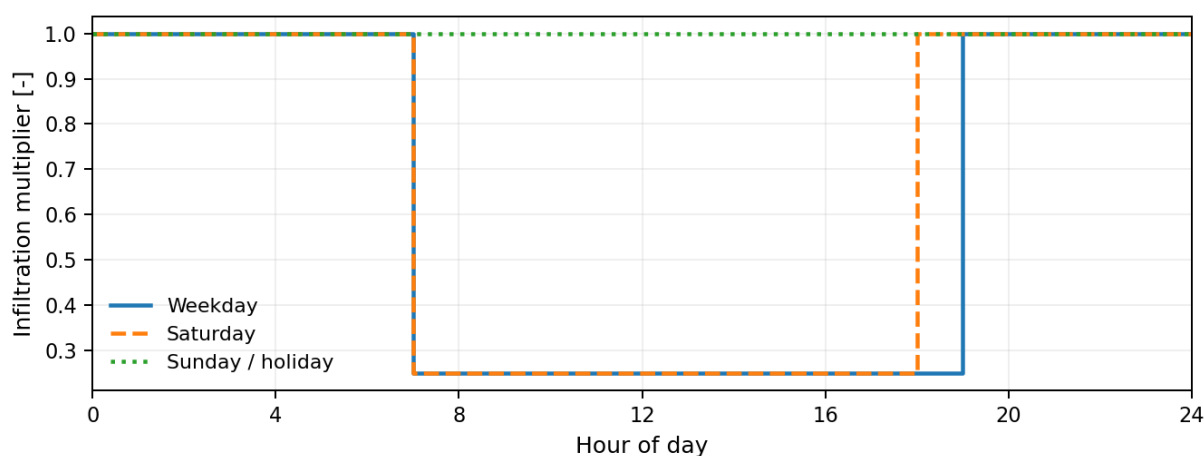


Figure 6-3. INFIL-SCH weekday, Saturday and Sunday profiles; the design-flow coefficients are separate from this multiplier. [1]

Source: [1, L5997–L6051]. The coefficient tuple is constant = 0, temperature = 0, wind = 0.224, and wind² = 0. The mechanical-ventilation outdoor-air request in Section 6.7 is an additional, different input. The exact INFIL-SCH day-type profiles appear in Appendix A.5.

6.10 HVAC Sizing and Performance Curves

Table 6-12. HVAC sizing inputs

Sizing input	Value	IDF location
Global heating / cooling factors	1.2 / 2.0	Sizing:Parameters
Explicit zone heating / cooling factors	1.2 / 1.2 in each of the three zone objects	Sizing:Zone
Zone cooling / heating supply-temperature method	SupplyAirTemperature / SupplyAirTemperature	Sizing:Zone
Zone cooling / heating design supply temperature	12.8 / 40°C	Sizing:Zone
Zone cooling / heating design supply humidity ratio	0.0085 / 0.008 kg/kg	Sizing:Zone
Zone dedicated outdoor-air system	No	Sizing:Zone
Zone load sizing method	Sensible Load Only No Latent Load	Sizing:Zone

Sizing input	Value	IDF location
Central load type / zone-sum method	Sensible / NonCoincident	Sizing:System
Central heating maximum airflow ratio	1.0	Sizing:System
Preheat design temperature / humidity ratio	7°C / 0.008 kg/kg	Sizing:System
Precool design temperature / humidity ratio	12.8°C / 0.008 kg/kg	Sizing:System
Central cooling / heating design supply temperature	12.8 / 17.7°C	Sizing:System
Central cooling / heating design humidity ratio	0.0085 / 0.008 kg/kg	Sizing:System
100% outdoor air: cooling / heating	No / No	Sizing:System
System design outdoor airflow	Autosize	Sizing:System
Cooling / heating design capacity	Autosize / Autosize	Sizing:System
Cooling / heating capacity method	CoolingDesignCapacity / HeatingDesignCapacity	Sizing:System
Central cooling capacity control	OnOff	Sizing:System

Global factors and explicit zone factors are separate inputs and are not multiplied together in this documentation. Alternative airflow/capacity sizing fields that are empty remain blank; the exact fields are listed in the register. The seven design-day definitions appear in Section 2.3. Reported capacity and airflow results are not substituted into Autosize fields. [1]

Table 6-13. The seven connected DX performance curves

Curve suffix	Form	Coefficients in entered order	Input domains
PLF	Quadratic	0.85, 0.15, 0	min x: 0; max x: 1
Cap-FF	Cubic	0.47278589, 1.2433415, -1.0387055, 0.32257813	min x: 0.5; max x: 1.5
EIR-FF	Cubic	1.0079484, 0.34544129, -0.6922891, 0.33889943	min x: 0.5; max x: 1.5
Cap-FT	Biquadratic	0.476428, 0.0401147, 0.000226411, -0.000827136, -0.000073224, -0.000446278	min x: 12.77778; max x: 23.88889; min y: 23.88889; max y: 46.11111
EIR-FT	Biquadratic	0.632475, -0.0121321, 0.000507773, 0.0155377, 0.00027284, -0.000679201	min x: 12.77778; max x: 23.88889; min y: 23.88889; max y: 46.11111
Low Cap-FT	Biquadratic	0.476428, 0.0401147, 0.000226411, -0.000827136, -0.000073224, -0.000446278	min x: 12.77778; max x: 23.88889; min y: 23.88889; max y: 46.11111
Low EIR-FT	Biquadratic	0.774645, -0.0343731, 0.000783173, 0.0146596, 0.000488851, -0.000752036	min x: 12.77778; max x: 23.88889; min y: 23.88889; max y: 46.11111

Source: [1, L6824–L6934]. The cubic records use Dimensionless input/output types; the biquadratic records use Temperature for x and y and Dimensionless output. Their minimum/maximum output-limit fields are blank. The quadratic record ends after its x bounds. Curve use is specified by the coil's name references. The fitted provenance of these curves is not established as measured HIBO equipment performance.

Four additional Battery-named curve records remain in the IDF but no battery or generation system references them. Their numerical definitions are retained in Appendix B.2 as unconnected input records; their presence does not mean battery operation is modeled. [1]

7. Simulation Results

The annual results cover 8,760 hours of the Omaha TMY3 simulation. Table 7-1 lists electricity by end use; Figure 7-1 shows the corresponding energy-use intensity (EUI). [5]

EUI is annual site energy divided by the modeled floor area of 67.78217 m², including MEP and excluding the duplicate plenum footprint. All reported end uses are electric. Values are shown in kWh/year, kWh/(m² yr), and kBtu/(ft² yr). [1,5]

Table 7-1. Annual simulation results - HIBO, Omaha TMY3

End use	Annual electricity [kWh/yr]	Site EUI [kWh/(m ² yr)]	Site EUI [kBtu/(ft ² yr)]
Interior equipment (electric)	944	13.93	4.42
Interior lighting (electric)	878	12.95	4.11
Exterior lighting (electric)	581	8.57	2.72
Heating incl. terminal reheat (electric)	2,092	30.86	9.78
Cooling (electric)	2,039	30.08	9.54
Fans (electric)	4,831	71.27	22.59
Total	11,366	167.69	53.16

Source: [1,5]. End-use values are read from the annual report at 0.01 GJ precision; total electricity is reported at 0.001 GJ. Converted kWh and EUI are approximate, and independently rounded entries may not sum exactly to the total.

Heating includes both the central electric heater and terminal electric reheat. No gas, elevator, hydronic-pump, service-water-heating, humidification, or dedicated refrigeration load is modeled. [1,5]

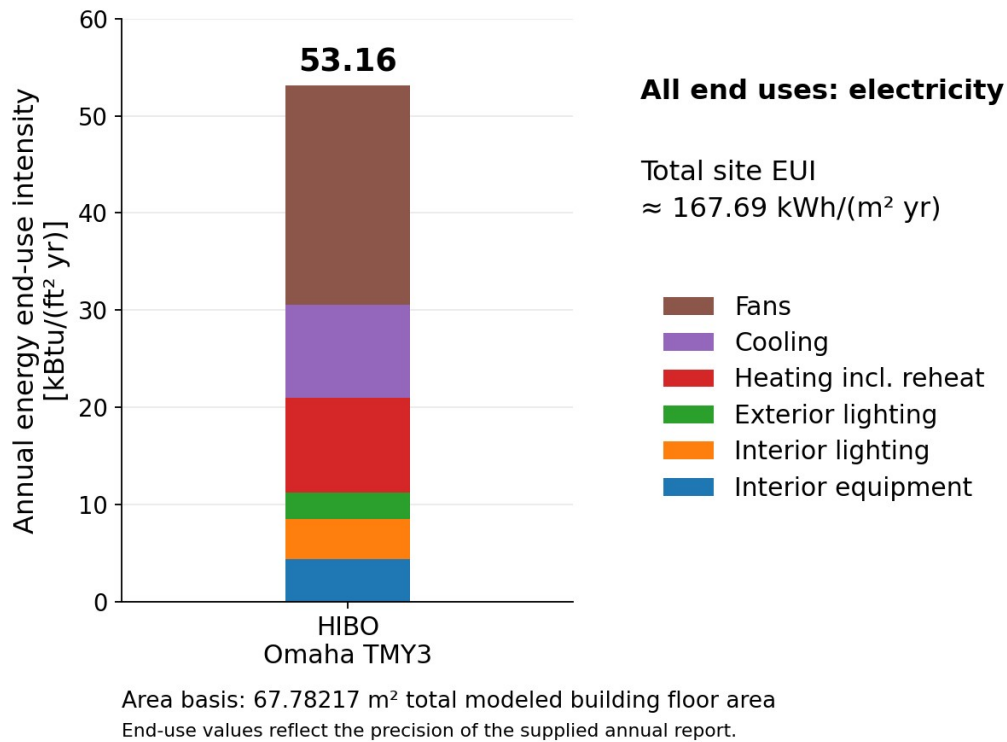


Figure 7-1. Annual site-energy end-use intensity for HIBO, Omaha TMY3. [5]

Fans are the largest end use, accounting for approximately 42.5% of annual electricity. Cooling accounts for 17.9%, and central heating plus terminal reheat for 18.4%. Heating, cooling, and fans together account for approximately 78.8%; interior equipment and interior/exterior lighting account for the remaining 21.2%. [5]

Total site electricity is approximately 11,366 kWh/year. Total site EUI is approximately 167.69 kWh/(m² yr), equivalent to 53.16 kBtu/(ft² yr). The supplied annual table does not separate central-heater and terminal-reheat electricity. [5]

The supplied run completed with three warnings and no severe errors. Occupied heating setpoint-not-met time is 53.83 facility-hours; occupied cooling setpoint-not-met time is 0.00 hours. These are simulated results, not measured-building performance. [5,6]

8. References

- [1] Bak, J. HIBO_09282026_JB.idf. User-supplied EnergyPlus input snapshot, 28 September 2026. Canonical file in model/. All input line references in this report refer to this unchanged file.
- [2] USA_NE_Omaha-Eppley.Airfield.725500_TMY3.epw. User-supplied weather file. LOCATION: Omaha Eppley Airfield; TMY3; WMO 725500. Header identifies NREL TMY Data Set (2008), generally 1973–2005 period of record. Included in weather/.
- [3] DLR Group / University of Nebraska–Omaha. Human-Centered Integrated Building Operation Lab, 95% Review Package, 24 February 2022. File: UNO-HCIBO-Lab_95%CheckSet_220223 - UNOReview.pdf. Principal sheets: A0.1, A1.1, A1.2, M1.1. This is design-intent evidence; reproduction restrictions are stated on the drawing sheets.
- [4] AAON, Inc. RTU-1 nameplate photograph, IMG_4634.jpeg, supplied by the user. Electric heat maximum 15 kW; field note reconfiguring unit from 6 ton to 4 ton; maximum outlet-air temperature 200°F. Photograph is equipment-rating evidence, not a full control-sequence document.
- [5] EnergyPlus. HIBO_09282026_JBTable.html and HIBO_09282026_JBZsz.csv. Supplied outputs from EnergyPlus 22.2.0-c249759bad, simulation timestamp 28 September 2026, 14:03:43. Table.html contains annual summaries; Zsz.csv contains zone-sizing design-day profiles. Both are included in reference/.
- [6] EnergyPlus. HIBO_09282026_JB.err. Supplied error log for the same run. A local-path-redacted copy is included in reference/.
- [7] Bonnema, E.; Leach, M.; Pless, S. Technical Support Document: Development of the Advanced Energy Design Guide for Large Hospitals – 50% Energy Savings. NREL/TP-5500-52588, June 2013. Sections 3.2–3.3 and Appendix C provide the model-description example; Section 3.3.8, pp. 57–58, provides the annual end-use table and stacked EUI format.

Material-property references recorded in the IDF:

- ASHRAE HOF F08 generic steel.
- CMHA TEK 06-02C, derived t/R without films.
- CMHA TEK 06-16A heat capacity, derived bulk-effective density.
- A0.1 R-7.5 equivalent / DuPont THERMAX R-7 per inch.
- EN 12524:2000, Table 1 EPDM reference.
- EnergyPlus 22.2 DOE reference: Metal Decking.
- EnergyPlus 22.2 Basement material reference: gravel.
- Armstrong ULTIMA High NRC 1940 reference: 7/8 in.
- NIST WebBook aluminum at 25°C; rounded.
- J+J Kinetex 1840 reference: nominal 0.205 in.

Appendix A. Schedule Tables

All twelve Schedule:Compact objects are shown below. They apply through December 31. Each row gives the value in the interval ending at the stated Until time; identical adjacent intervals and identical day-type profiles may be combined for readability. Full-precision fields and original selectors remain in the complete input register. The schedules are inputs, not measured BAS trends. [1]

A.1 Occupancy Schedules

Table A-1. OCCUPY-1

Day type(s)	Interval	Value [fraction]
Weekdays	00:00–06:00	0
Weekdays	06:00–07:00	0.11
Weekdays	07:00–08:00	0.21
Weekdays	08:00–12:00	1
Weekdays	12:00–13:00	0.53
Weekdays	13:00–17:00	1
Weekdays	17:00–18:00	0.32
Weekdays	18:00–22:00	0.11
Weekdays	22:00–23:00	0.05
Weekdays	23:00–24:00	0
Weekends; Holiday; WinterDesignDay; CustomDay1; CustomDay2	00:00–24:00	0
SummerDesignDay	00:00–24:00	1

Type limits: Fraction. [1, L508–L550].

Table A-2. ActSchd

Day type(s)	Interval	Value [W/person]
AllDays	00:00–24:00	117.239998

Type limits: Any Number. [1, L602–L608].

A.2 Thermostat Set Point Schedules

Table A-3. HTGSETP_SCH_NO_OPTIMUM

Day type(s)	Interval	Value [°C]
WeekDays	00:00–06:00	15.56
WeekDays	06:00–19:00	21.11
WeekDays	19:00–24:00	15.56
Weekends; Holiday; SummerDesignDay; CustomDay1; CustomDay2	00:00–24:00	15.56
WinterDesignDay	00:00–24:00	21.11

Type limits: Temperature. [1, L394–L424].

Table A-4. CLGSETP_SCH_NO_OPTIMUM

Day type(s)	Interval	Value [°C]
WeekDays	00:00–06:00	29.44
WeekDays	06:00–18:00	23.89
WeekDays	18:00–24:00	29.44
Weekends; Holiday; WinterDesignDay; CustomDay1; CustomDay2	00:00–24:00	29.44
SummerDesignDay	00:00–24:00	23.89

Type limits: Temperature. [1, L426–L456].

A.3 Lighting Schedules

Table A-5. Itg_sch_office

Day type(s)	Interval	Value [fraction]
SummerDesignDay	00:00–24:00	1
WinterDesignDay	00:00–24:00	0
Weekdays	00:00–05:00	0.114132
Weekdays	05:00–07:00	0.145835
Weekdays	07:00–08:00	0.266306
Weekdays	08:00–12:00	0.570652
Weekdays	12:00–13:00	0.507247
Weekdays	13:00–17:00	0.570652
Weekdays	17:00–18:00	0.386777
Weekdays	18:00–20:00	0.266306
Weekdays	20:00–22:00	0.202901
Weekdays	22:00–23:00	0.145835
Weekdays	23:00–24:00	0.114132
Saturday; AllOtherDays	00:00–24:00	0.114132

Type limits: Fraction. [1, L458–L496].

Table A-6. Exterior_lighting_schedule_a

Day type(s)	Interval	Value [fraction]
AllDays	00:00–06:00	0
AllDays	06:00–24:00	1

Type limits: fraction. [1, L498–L506].

A.4 Plug and Process Load Schedules

Table A-7. EQUIP-1

Day type(s)	Interval	Value [fraction]
WeekDays SummerDesignDay CustomDay1 CustomDay2	00:00–08:00	0.02
WeekDays SummerDesignDay CustomDay1 CustomDay2	08:00–09:00	0.4
WeekDays SummerDesignDay CustomDay1 CustomDay2	09:00–14:00	0.9
WeekDays SummerDesignDay CustomDay1 CustomDay2	14:00–15:00	0.8
WeekDays SummerDesignDay CustomDay1 CustomDay2	15:00–16:00	0.7
WeekDays SummerDesignDay CustomDay1 CustomDay2	16:00–18:00	0.5

Day type(s)	Interval	Value [fraction]
WeekDays SummerDesignDay CustomDay1 CustomDay2	18:00–21:00	0.3
WeekDays SummerDesignDay CustomDay1 CustomDay2	21:00–24:00	0.02
Weekends WinterDesignDay Holiday	00:00–24:00	0.02

Type limits: Fraction. [1, L552–L575].

A.5 Infiltration and Outdoor-Air Schedules

Table A-8. INFIL-SCH

Day type(s)	Interval	Value [fraction]
Weekdays	00:00–07:00	1
Weekdays	07:00–19:00	0.25
Weekdays	19:00–24:00	1
Saturday	00:00–07:00	1
Saturday	07:00–18:00	0.25
Saturday	18:00–24:00	1
WinterDesignDay SummerDesignDay; Sunday Holidays AllOtherDays	00:00–24:00	1

Type limits: Fraction. [1, L577–L600].

Table A-9. Min OA Sched

Day type(s)	Interval	Value [fraction]
WeekDays CustomDay1 CustomDay2	00:00–06:00	0
WeekDays CustomDay1 CustomDay2	06:00–23:00	1
WeekDays CustomDay1 CustomDay2	23:00–24:00	0
WeekEnds Holiday	00:00–24:00	0
SummerDesignDay; WinterDesignDay	00:00–24:00	1

Type limits: Fraction. [1, L627–L646].

A.6 Availability and Constant Control Schedules

Table A-10. AvailSched_1

Day type(s)	Interval	Value [on/off]
WeekDays CustomDay1 CustomDay2; WeekEnds Holiday; SummerDesignDay; WinterDesignDay	00:00–24:00	1

Type limits: On/Off. [1, L610–L625].

Table A-11. HVACTemplate-Always 4

Day type(s)	Interval	Value [control code]
AllDays	00:00–24:00	4

Type limits: Any Number. [1, L648–L654].

Table A-12. HVACTemplate-Always 1

Day type(s)	Interval	Value [availability value]
AllDays	00:00–24:00	1

Type limits: Any Number. [1, L656–L662].

A.7 Schedule Type Limits

Table A-13. All six ScheduleTypeLimits records

Name	Lower bound	Upper bound	Numeric type	Unit type field
Any Number	Not entered	Not entered	Not entered	Not entered
Fraction	0	1	Continuous	Not entered
Temperature	-60	200	Continuous	Not entered
Control Type	0	4	Discrete	Not entered
On/Off	0	1	Discrete	Not entered
FlowRate	0	10	Continuous	Not entered

Source: [1, L358–L389]. FlowRate is retained as a type-limit record even though none of the twelve Schedule:Compact objects uses it. The six unit-type fields are omitted in the entered records; the quantity meanings used in this report follow each schedule’s consuming object, not an invented unit-type entry.

Appendix B. Additional Input Settings

B.1 Air-System Connections

Table B-1. Common HVAC connection sequence

Position / object	Inlet → outlet / reference
Common return plenum	Three main-zone return nodes → DXVAV Sys 1 Return Air Outlet
Return fan	DXVAV Sys 1 Air Loop Inlet → DXVAV Sys 1 Return Fan Outlet
Outdoor/return mixer	Return Fan Outlet plus outdoor-air node → DXVAV Sys 1 Mixed Air Outlet; relief node separate
Two-speed DX coil	Mixed Air Outlet → DXVAV Sys 1 Cooling Coil Outlet
Central heater	Cooling Coil Outlet → DXVAV Sys 1 Heating Coil Outlet
Supply fan	Heating Coil Outlet → DXVAV Sys 1 Supply Fan Outlet
Supply path and splitter	DXVAV Sys 1 Supply Path Inlet → each zone Zone Equip Inlet
Each terminal	Zone Equip Inlet → Damper Outlet → Supply Inlet
Each zone equipment connection	Supply Inlet / Zone Air Node / Return Outlet

Source: [1, L6306–L6819]. Connection labels in this table abbreviate repeated prefixes. The field register preserves full node names, side-specific AirLoopHVAC nodes, equipment sequence numbers, and the fan and mixed-air setpoint-manager references.

Table B-2. Air-loop support-object settings

Support object	Entered settings
ZoneHVAC:EquipmentList	SequentialLoad; one AirDistributionUnit per main zone; heating and cooling sequence = 1
ZoneHVAC:AirDistributionUnit	One VAV terminal reference per zone; additional leakage fields omitted
OutdoorAir:Node	DXVAV Sys 1 Outside Air Inlet; height and trailing fields omitted
AirLoopHVAC:ControllerList	One Controller:OutdoorAir entry in the OA system
AirLoopHVAC:OutdoorAirSystem:EquipmentList	One OutdoorAir:Mixer entry
SupplyPath / ReturnPath	One ZoneSplitter / one ReturnPlenum
NodeList	SAT Setpoint Nodes: supply-path inlet and supply-fan outlet; Mixed Air Nodes: cooling-coil outlet and mixed-air outlet
SetpointManager:MixedAir	Cooling manager reference = supply-fan outlet; heating manager reference = supply-path inlet. Both use fan inlet = heating-coil outlet and fan outlet = supply-fan outlet.

The exact fields, including omitted availability-manager lists and controller-list blanks, appear in the complete register. No connector, plant loop, air-distribution leakage model, humidity controller, or separate external control interface is inferred merely from a generic HVAC object name. [1]

B.2 Unconnected Curve Records

Table B-3. BatteryChargeCurve (Curve:RectangularHyperbola2)

Field	Entered value
Coefficient1 C1	-.2765
Coefficient2 C2	-93.27
Coefficient3 C3	0.0068

Field	Entered value
Minimum Value of x	0
Maximum Value of x	1
Minimum Curve Output	-100
Maximum Curve Output	100

Source: [1, L6939–L6947]. No active battery or generator object references this record.

Table B-4. BatteryDischargeCurve (Curve:RectangularHyperbola2)

Field	Entered value
Coefficient1 C1	0.0899
Coefficient2 C2	-98.24
Coefficient3 C3	-.0082
Minimum Value of x	0
Maximum Value of x	1
Minimum Curve Output	-100
Maximum Curve Output	100

Source: [1, L6949–L6957]. No active battery or generator object references this record.

Table B-5. old_BatteryCycleCurve (Curve:DoubleExponentialDecay)

Field	Entered value
Coefficient1 C1	1380
Coefficient2 C2	6834
Coefficient3 C3	-8.75
Coefficient4 C4	6747
Coefficient5 C5	-6.22
Minimum Value of x	0
Maximum Value of x	1

Source: [1, L6962–L6970]. No active battery or generator object references this record.

Table B-6. new_BatteryCycleCurve (Curve:DoubleExponentialDecay)

Field	Entered value
Coefficient1 C1	1380
Coefficient2 C2	34170
Coefficient3 C3	-1.75
Coefficient4 C4	33735
Coefficient5 C5	-1.244
Minimum Value of x	0
Maximum Value of x	1

Source: [1, L6972–L6980]. No active battery or generator object references this record.

B.3 Reporting and Diagnostic Inputs

The IDF requests the following summary reports, diagnostic dictionaries, and time-series outputs. [1]

Table B-7. Summary, dictionary, drawing, database and diagnostic requests

Class	Entered request
Output:VariableDictionary	Key Field: regular
Output:Surfaces:List	Report Type: Lines
Output:Surfaces:Drawing	Report Type: DXF
Output:Schedules	Key Field: Timestep
Output:Constructions	Details Type 1: Constructions; Details Type 2: Materials
Output:EnergyManagementSystem	Actuator Availability Dictionary Reporting: Verbose; Internal Variable Availability Dictionary Reporting: Verbose; EMS Runtime Language Debug Output Level: ErrorsOnly
Output:Table:SummaryReports	Report 1 Name: AllSummary
OutputControl:Table:Style	Column Separator: HTML
Output:SQLite	Option Type: SimpleAndTabular
Output:Diagnostics	Key 1: DisplayExtraWarnings; Key 2: DisplayAdvancedReportVariables

There are 127 Output:Variable records and 14 Output:Meter:MeterFileOnly records. Repeated requests are retained in the register instead of being silently deleted. Requesting a variable does not guarantee a populated physical output for every key; for example, a system node without a setpoint can report an unset marker. No separate measured-temperature record is implied. [1]

Table B-8. All meter-file-only requests

Meter name	Frequency
Electricity:Facility	Hourly
Electricity:Building	Hourly
InteriorLights:Electricity	Hourly
Electricity:HVAC	Hourly
Electricity:Facility	runperiod
Electricity:Building	runperiod
InteriorLights:Electricity	runperiod
Electricity:HVAC	runperiod
Heating:Electricity	Hourly
Cooling:Electricity	Hourly
Electricity:Zone:THERMAL ZONE: SPACE 103 LAB B	Hourly
Electricity:Zone:THERMAL ZONE: SPACE 102 - PLUS	Hourly
Electricity:Zone:THERMAL ZONE: SPACE 105 LAB A	Hourly
ElectricityNet:Facility	Hourly

Table B-9. All requested variable/key/frequency combinations (duplicate counts preserved)

Requested variable	Key / scope	Frequency / count
Zone Air Temperature	*	Timestep
Site Outdoor Air Relative Humidity	*	Timestep
Facility Total Electricity Demand Rate	Whole Building	Timestep
Site Outdoor Air Drybulb Temperature	*	Timestep
Zone Air Relative Humidity	*	Timestep

Requested variable	Key / scope	Frequency / count
System Node Temperature	*	Timestep × 2
System Node Setpoint Temperature	*	Timestep × 2
Site Direct Solar Radiation Rate per Area	*	Timestep
Facility Net Purchased Electricity Rate	*	Timestep
Cooling Coil Total Cooling Rate	DXVAV Sys 1 Cooling Coil	TimeStep
Cooling Coil Electricity Rate	DXVAV Sys 1 Cooling Coil	TimeStep
Heating Coil Heating Rate	DXVAV Sys 1 Heating Coil	TimeStep
Heating Coil Electricity Rate	DXVAV Sys 1 Heating Coil	TimeStep
Heating Coil Heating Rate	Thermal Zone: Space 103 LAB B Reheat Coil	TimeStep
Heating Coil Heating Rate	Thermal Zone: Space 102 - Plus Reheat Coil	TimeStep
Heating Coil Heating Rate	Thermal Zone: Space 105 LAB A Reheat Coil	TimeStep
Fan Electricity Rate	DXVAV Sys 1 Supply Fan	TimeStep
Fan Air Mass Flow Rate	DXVAV Sys 1 Supply Fan	TimeStep
Fan Electricity Rate	DXVAV Sys 1 Return Fan	TimeStep
System Node Temperature	DXVAV Sys 1 Mixed Air Outlet	TimeStep
System Node Temperature	DXVAV Sys 1 Cooling Coil Outlet	TimeStep
System Node Temperature	DXVAV Sys 1 Heating Coil Outlet	TimeStep
System Node Temperature	DXVAV Sys 1 Supply Fan Outlet	TimeStep
System Node Temperature	Thermal Zone: Space 103 LAB B Zone Equip Inlet	TimeStep
System Node Temperature	Thermal Zone: Space 102 - Plus Zone Equip Inlet	TimeStep
System Node Temperature	Thermal Zone: Space 105 LAB A Zone Equip Inlet	TimeStep
Air System Outdoor Air Economizer Status	DXVAV Sys 1	TimeStep
Air System Outdoor Air Flow Fraction	DXVAV Sys 1	TimeStep
Cooling Coil Sensible Cooling Rate	DXVAV Sys 1 Cooling Coil	TimeStep
Cooling Coil Latent Cooling Rate	DXVAV Sys 1 Cooling Coil	TimeStep
Cooling Coil Runtime Fraction	DXVAV Sys 1 Cooling Coil	TimeStep
System Node Mass Flow Rate	DXVAV Sys 1 Mixed Air Outlet	TimeStep
System Node Mass Flow Rate	DXVAV Sys 1 Supply Fan Outlet	TimeStep
System Node Humidity Ratio	DXVAV Sys 1 Mixed Air Outlet	TimeStep
System Node Humidity Ratio	DXVAV Sys 1 Supply Fan Outlet	TimeStep
System Node Setpoint Temperature	DXVAV Sys 1 Supply Fan Outlet	TimeStep
Zone Thermostat Heating Setpoint Temperature	Thermal Zone: Space 103 LAB B	TimeStep
Zone Thermostat Cooling Setpoint Temperature	Thermal Zone: Space 103 LAB B	TimeStep
Zone Predicted Sensible Load to Heating Setpoint Heat Transfer Rate	Thermal Zone: Space 103 LAB B	TimeStep
Zone Predicted Sensible Load to Cooling Setpoint Heat Transfer Rate	Thermal Zone: Space 103 LAB B	TimeStep

Requested variable	Key / scope	Frequency / count
Zone Air Terminal VAV Damper Position	Thermal Zone: Space 103 LAB B VAV Reheat	TimeStep
Zone Thermostat Heating Setpoint Temperature	Thermal Zone: Space 102 - Plus	TimeStep
Zone Thermostat Cooling Setpoint Temperature	Thermal Zone: Space 102 - Plus	TimeStep
Zone Predicted Sensible Load to Heating Setpoint Heat Transfer Rate	Thermal Zone: Space 102 - Plus	TimeStep
Zone Predicted Sensible Load to Cooling Setpoint Heat Transfer Rate	Thermal Zone: Space 102 - Plus	TimeStep
Zone Air Terminal VAV Damper Position	Thermal Zone: Space 102 - Plus VAV Reheat	TimeStep
Zone Thermostat Heating Setpoint Temperature	Thermal Zone: Space 105 LAB A	TimeStep
Zone Thermostat Cooling Setpoint Temperature	Thermal Zone: Space 105 LAB A	TimeStep
Zone Predicted Sensible Load to Heating Setpoint Heat Transfer Rate	Thermal Zone: Space 105 LAB A	TimeStep
Zone Predicted Sensible Load to Cooling Setpoint Heat Transfer Rate	Thermal Zone: Space 105 LAB A	TimeStep
Zone Air Terminal VAV Damper Position	Thermal Zone: Space 105 LAB A VAV Reheat	TimeStep
Site Day Type Index	*	Timestep
Zone Windows Total Transmitted Solar Radiation Rate	*	Timestep
Zone Exterior Windows Total Transmitted Beam Solar Radiation Rate	*	Timestep
Zone Exterior Windows Total Transmitted Diffuse Solar Radiation Rate	*	Timestep
Zone Total Internal Convective Heating Rate	*	Timestep
Zone Total Internal Radiant Heating Rate	*	Timestep
Zone Total Internal Visible Radiation Heating Rate	*	Timestep
Zone Total Internal Latent Gain Rate	*	Timestep
Zone Total Internal Total Heating Rate	*	Timestep
Zone People Occupant Count	*	Timestep
Zone People Convective Heating Rate	*	Timestep
Zone People Radiant Heating Rate	*	Timestep
Zone People Sensible Heating Rate	*	Timestep
Zone People Latent Gain Rate	*	Timestep
Zone People Total Heating Rate	*	Timestep
Zone Electric Equipment Radiant Heating Rate	*	Timestep
Zone Electric Equipment Convective Heating Rate	*	Timestep
Zone Electric Equipment Latent Gain Rate	*	Timestep
Zone Electric Equipment Lost Heat Rate	*	Timestep
Zone Electric Equipment Total Heating Rate	*	Timestep
Zone Infiltration Sensible Heat Gain Energy	*	Timestep
Zone Infiltration Sensible Heat Loss Energy	*	Timestep
Zone Infiltration Latent Heat Gain Energy	*	Timestep
Zone Infiltration Latent Heat Loss Energy	*	Timestep

Requested variable	Key / scope	Frequency / count
Zone Infiltration Total Heat Gain Energy	*	Timestep
Zone Infiltration Total Heat Loss Energy	*	Timestep
Zone Infiltration Mass Flow Rate	*	Timestep
Zone Infiltration Current Density Volume Flow Rate	*	Timestep
Zone Infiltration Air Change Rate	*	Timestep
HVAC System Solver Iteration Count	*	Timestep
Air System Solver Iteration Count	*	Timestep
Air System Outdoor Air Mass Flow Rate	DXVAV Sys 1	Timestep
Air System Mixed Air Mass Flow Rate	DXVAV Sys 1	Timestep
Air System Outdoor Air Minimum Flow Fraction	DXVAV Sys 1	Timestep
Zone Air Terminal Outdoor Air Volume Flow Rate	*	Timestep
Zone Air Terminal Minimum Air Flow Fraction	*	Timestep
Schedule Value	Min OA Sched	Timestep
System Node Mass Flow Rate	Thermal Zone: Space 103 LAB B Supply Inlet	Timestep
System Node Mass Flow Rate	Thermal Zone: Space 102 - Plus Supply Inlet	Timestep
System Node Mass Flow Rate	Thermal Zone: Space 105 LAB A Supply Inlet	Timestep
Zone Heating Setpoint Not Met Time	*	Timestep
Zone Cooling Setpoint Not Met Time	*	Timestep
Zone Heating Setpoint Not Met While Occupied Time	*	Timestep
Zone Cooling Setpoint Not Met While Occupied Time	*	Timestep
Air System Outdoor Air Mechanical Ventilation Requested Mass Flow Rate	DXVAV Sys 1	Timestep
System Node Standard Density Volume Flow Rate	Thermal Zone: Space 103 LAB B Supply Inlet	Timestep
System Node Standard Density Volume Flow Rate	Thermal Zone: Space 102 - Plus Supply Inlet	Timestep
System Node Standard Density Volume Flow Rate	Thermal Zone: Space 105 LAB A Supply Inlet	Timestep
System Node Standard Density Volume Flow Rate	DXVAV Sys 1 Supply Fan Outlet	Timestep
Site Ground Temperature	*	Timestep
Surface Inside Face Temperature	Surface 1	Timestep
Surface Outside Face Temperature	Surface 1	Timestep
Surface Inside Face Conduction Heat Transfer Rate	Surface 1	Timestep
Surface Inside Face Convection Heat Gain Rate	Surface 1	Timestep
Surface Inside Face Temperature	Surface 9	Timestep
Surface Outside Face Temperature	Surface 9	Timestep
Surface Inside Face Conduction Heat Transfer Rate	Surface 9	Timestep
Surface Inside Face Convection Heat Gain Rate	Surface 9	Timestep
Surface Inside Face Temperature	Surface 15	Timestep
Surface Outside Face Temperature	Surface 15	Timestep
Surface Inside Face Conduction Heat Transfer Rate	Surface 15	Timestep

Requested variable	Key / scope	Frequency / count
Surface Inside Face Convection Heat Gain Rate	Surface 15	Timestep
Surface Inside Face Temperature	Surface 23	Timestep
Surface Outside Face Temperature	Surface 23	Timestep
Surface Inside Face Conduction Heat Transfer Rate	Surface 23	Timestep
Surface Inside Face Convection Heat Gain Rate	Surface 23	Timestep
Surface Inside Face Temperature	Surface 33	Timestep
Surface Outside Face Temperature	Surface 33	Timestep
Surface Inside Face Conduction Heat Transfer Rate	Surface 33	Timestep
Surface Inside Face Convection Heat Gain Rate	Surface 33	Timestep
Surface Inside Face Temperature	Surface 40	Timestep
Surface Outside Face Temperature	Surface 40	Timestep
Surface Inside Face Conduction Heat Transfer Rate	Surface 40	Timestep
Surface Inside Face Convection Heat Gain Rate	Surface 40	Timestep

Appendix C. Abbreviations and Symbols

Term	Meaning
AHU / RTU	Air-handling unit / rooftop unit
BAS / BMS	Building automation system / building management system
CFS / CMU	Cold-formed steel / concrete masonry unit
COP	Coefficient of performance
CTF	Conduction transfer function
CTG / CTIG / ITSG	Clear tempered glass / clear tempered insulating glass / insulated tempered spandrel glass
DAT / SAT	Discharge-air temperature / supply-air temperature
DX	Direct expansion
EPW / IDF	EnergyPlus weather file / input data file
HVAC	Heating, ventilation, and air conditioning
OA / RA / SA	Outdoor / return / supply air
TMY3	Typical Meteorological Year, third-generation data set
VAV	Variable air volume
R / C_A	Layer thermal resistance [$\text{m}^2 \text{ K/W}$] / areal heat capacity [$\text{kJ}/(\text{m}^2 \text{ K})$]
t / k / ρ / c_p	Thickness / thermal conductivity / density / specific heat
Autosize / Autocalculate	Literal sizing or calculation directives retained in the input
Blank	An explicitly empty field; not a numeric zero and not a measured value
EUI	Energy-use intensity; annual site energy per unit floor area